Rubric physics research report

Each criterion receives a grade from 1 to 10.

1. Theory and research question Correct physical theory, accurately stated, with a testable question and hypothesis.

2. Method and measurement uncertainty Replicable measurement set-up, justified data selection, uncertainties correctly determined and propagated.

3. Calculations Formulas applied consistently and correctly, with the right units. A unit error (for example degrees Celsius where kelvin is required) is a serious error.

4. Interpretation and discussion Results critically tested against theory. When a result is impossible, the student checks their own calculation before blaming the measuring equipment.

5. Conclusion Answers the research question, tests the hypothesis, and does not go beyond what the data allow.

6. Scientific reporting Clear structure, correct notation of quantities and uncertainties, careful use of language.